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Decomposition-based hybrid methods employing statistical, machine learning, and deep learning models for crude oil price forecasting

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governments, policymakers, and stakeholders to make well-informed decisions and effectively mitigate risks. The decomposition-based hybrid models have been showing promising COP forecasting accuracy than other time series forecasting methods. Despite this fact, no systematic study has been conducted to evaluate the true potential of different decomposition-based hybrid methods employing different forecasting models to forecast the COP. Therefore, a hybrid modeling framework is developed by combining efficient decomposition techniques, namely empirical mode decomposition (EMD), ensemble EMD (EEMD), complete EEMD with adaptive noise (CEEMDAN), and variational mode decomposition (VMD) with seven statistical models, fourteen machine learning (ML) models, and six deep learning (DL) models. Further, a systematic study is conducted on the resulting decomposition-based hybrid models to find the best hybrid model for COP forecasting. Three distinct train-test data splits are employed to ensure a reliable evaluation of the models using four performance metrics. Extensive statistical analysis is conducted to identify the optimal combination of the decomposition technique and forecasting model for

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Data availability

The authors confirm that the data supporting the findings of this study are available. Raw data that support the findings of this study are available from the corresponding author upon reasonable request.

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